MF 281: Probability Theory

Week 5: Discrete random variables

Attaching a number to every outcome

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Lecture notes for students of MF 281 Probability Theory.

Week 5: Discrete random variables

Learning Objectives

By the end of this week you should be able to:

- define a random variable as a function on the sample space, and read the notation $\{X = x\}$ correctly;
- construct a probability mass function from a sample space, and check it is valid;
- draw and use a cumulative distribution function, and recover the mass function from it;
- find the distribution of a function of a random variable;
- recognise an indicator variable and say what it is for.

5.1 From events to numbers

So far a question was an event: did the system fail, was the bolt defective. Most real questions are quantitative instead: how many bolts were defective, how long until the system failed, what did the shift cost.

Definition 5.1 (Random variable)**.** A random variable is a function $X : \Omega \to \mathbb{R}$. For each outcome ω, it returns a number $X(\omega)$.

Two warnings about the name, which is badly chosen and two centuries old. A random variable is not random, and it is not a variable: it is an ordinary deterministic function. The randomness lives in which ω occurs; X merely reports a number once it has.

The notation takes a moment to get used to:

$$\{X = x\} \text{ means } \{\omega \in \Omega : X(\omega) = x\},$$

an *event*, and $P(X = x)$ is its probability. Similarly $\{X \leq x\}$, $\{a < X \leq b\}$ and so on are events, so everything from Weeks 2 to 4 applies to them unchanged.

Example 1: three coin tosses

Toss a fair coin three times, so Ω has the eight sequences HHH to TTT. Let X be the number of heads.

Solution

X is the function $X(HHH) = 3$, $X(HHT) = X(HTH) = X(THH) = 2$, and so on. Then

$$\{X = 2\} = \{HHT, HTH, THH\}, \qquad P(X = 2) = \tfrac{3}{8}.$$

Collecting all four values:

$$P(X = 0) = \tfrac{1}{8}, \quad P(X = 1) = \tfrac{3}{8}, \quad P(X = 2) = \tfrac{3}{8}, \quad P(X = 3) = \tfrac{1}{8}.$$

Eight outcomes have been compressed into four numbers, and nothing about the ordering of the tosses survives. That compression is the point of a random variable: it keeps what the question needs and discards the rest.

5.2 The probability mass function

Definition 5.2 (Discrete random variable and pmf)**.** X is discrete if it takes values in a finite or countable set. Its probability mass function is

$$p_X(x) = P(X = x).$$

Proposition 5.1. *A function p is the pmf of some discrete random variable if and only if $p(x) \geq 0$ for all x and $\sum_x p(x) = 1$.*

Both conditions come straight from the axioms: non-negativity from the first, and the sum from the fact that the events $\{X = x\}$ are disjoint and their union is Ω.

Example 2: the sum of two dice

Roll two fair dice and let S be the total.

s	2	3	4	5	6	7	8	9	10	11	12
$p_S(s)$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$

Solution

The 36 equally likely pairs are counted into eleven values. The masses sum to $36/36 = 1$, as they must, and $P(S = 7) = 6/36 = 1/6$ is the largest because seven has the most ways of happening.

Notice that S is not uniform even though the underlying outcomes are. A random variable can concentrate probability simply by mapping many outcomes to one value.

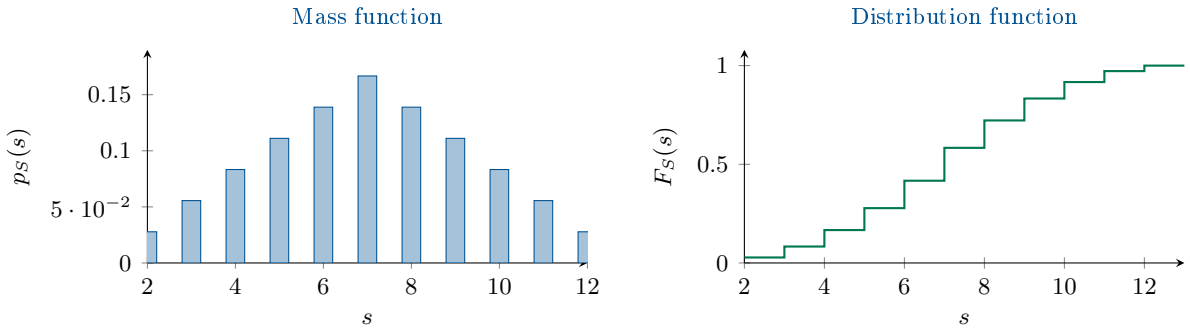


Figure 5.1: The same distribution twice. The mass function says how much probability sits at each value; the distribution function accumulates it from the left.

5.3 The cumulative distribution function

Definition 5.3 (CDF). $F_X(x) = P(X \leq x)$, defined for every real x, not only the values X can take.

For a discrete variable, F_X is a staircase: flat between the possible values, with a jump of exactly $p_X(x)$ at each value x. Three properties, all immediate from the axioms:

- F_X is non-decreasing, by monotonicity;
- $F_X(x) \to 0$ as $x \to -\infty$ and $F_X(x) \to 1$ as $x \to +\infty$;
- F_X is right-continuous, because the inequality is $\leq$ rather than $<$.

The two descriptions carry the same information:

$$F_X(x) = \sum_{t \leq x} p_X(t), \qquad p_X(x) = F_X(x) - F_X(x^-),$$

where $F_X(x^-)$ is the value just to the left, so the mass at x is the size of the jump.

Example 3: reading the staircase

For the dice sum of Example 2:

$$F_S(4) = \tfrac{1+2+3}{36} = \tfrac{1}{6} = 0.1667, \qquad F_S(7) = \tfrac{21}{36} = \tfrac{7}{12} = 0.5833, \qquad F_S(9) = \tfrac{30}{36} = \tfrac{5}{6}.$$

Also $F_S(4.7) = F_S(4) = 1/6$, because no value lies between 4 and 4.7.

Intervals from the CDF

$$P(4 < S \le 9) = F_S(9) - F_S(4) = \tfrac{5}{6} - \tfrac{1}{6} = \tfrac{2}{3}.$$

Mind the endpoints: $P(4 \le S \le 9)$ would be $F_S(9) - F_S(3) = 5/6 - 1/12 = 3/4$. For discrete variables the difference between $<$ and $\le$ is real, unlike the continuous case in Week 9.

5.4 Functions of a random variable

If X is a random variable and g a function, then $Y = g(X)$ is a random variable too, with

$$p_Y(y) = \sum_{x\,:\,g(x)=y} p_X(x).$$

Values of x that g sends to the same place have their masses added.

Example 4: squaring collapses values

From Example 1, let X be the number of heads in three tosses, and let $Y = (X-1)^2$, say a cost that rises with any departure from one head.

Solution

The map sends $0 \mapsto 1$, $1 \mapsto 0$, $2 \mapsto 1$ and $3 \mapsto 4$, so

$$p_Y(0) = p_X(1) = \tfrac{3}{8}, \qquad p_Y(1) = p_X(0) + p_X(2) = \tfrac{1}{8} + \tfrac{3}{8} = \tfrac{1}{2}, \qquad p_Y(4) = p_X(3) = \tfrac{1}{8}.$$

The masses sum to $3/8 + 1/2 + 1/8 = 1$. Y takes three values where X took four, because g is not injective, and the probabilities of the merged values add.

5.5 Two variables worth knowing already

5.5.1 The indicator

For an event A, define $\mathbf{1}_A(\omega) = 1$ if $\omega \in A$ and 0 otherwise. Its pmf is $P(\mathbf{1}_A = 1) = P(A)$ and $P(\mathbf{1}_A = 0) = 1 - P(A)$.

Indicators are the bridge between events and numbers: counting how many of several events occur is the same as adding their indicators, which is how many apparently hard problems become easy in Week 7.

5.5.2 A count of successes

Example 5: defective bolts

Four bolts are drawn independently, each defective with probability 0.2. Let X be the number defective.

x	0	1	2	3	4
$p_X(x)$	0.4096	0.4096	0.1536	0.0256	0.0016
$F_X(x)$	0.4096	0.8192	0.9728	0.9984	1.0000

Solution

Each entry is $\binom{4}{x}(0.2)^x(0.8)^{4-x}$, from Week 4's independent trials. The masses sum to one, and the CDF reaches one at $x = 4$, two checks worth making every time you tabulate a distribution. From the table, $P(X \geq 2) = 1 - F_X(1) = 1 - 0.8192 = 0.1808$, which is quicker than adding three terms. This is the binomial distribution, and Week 6 studies it properly.

Example 6: waiting for the first defect

Now inspect bolts one at a time until the first defective one appears, with the same $p = 0.2$. Let N be the number inspected.

Solution

The first $k-1$ must be sound and the kth defective, so by independence

$$p_N(k) = (0.8)^{k-1}(0.2), \qquad k = 1, 2, 3, \ldots$$

This is a countably infinite sample space, handled by Week 2's axioms. The masses sum to one as a geometric series. Then

$$P(N \leq 3) = 1 - (0.8)^3 = 0.488, \qquad P(N > 5) = (0.8)^5 = 0.3277.$$

The second follows from the useful identity $P(N > k) = (1-p)^k$: surviving k inspections means k sound bolts in a row.

5.6 Summary

Week 5 in six lines

- A random variable is a function on Ω; the randomness is in ω.
- $\{X = x\}$ is an event, so every earlier rule still applies.
- A pmf is non-negative and sums to one; check both before using it.
- The CDF accumulates mass from the left and jumps by $p_X(x)$ at each value.
- For $Y = g(X)$, add the masses of all x mapping to the same y.
- Indicators turn events into numbers, which is how counting becomes algebra.

Exercises

1. A fair coin is tossed twice. Let X be the number of heads and Y the number of changes between consecutive tosses. Write both as functions on Ω, and give both mass functions.

2. For the dice sum S, find $P(S \geq 10)$, $F_S(6)$ and $P(6 < S \leq 10)$.

3. A pmf is proposed as $p(x) = c/x$ for $x = 1, 2, 3, 4$. Find c, then $P(X \leq 2)$.

4. Let X have $p_X(-2) = 0.2$, $p_X(-1) = 0.1$, $p_X(1) = 0.3$, $p_X(2) = 0.4$. Find the pmf of $Y = X^2$ and of $Z = |X| + 1$.

5. Sketch the CDF of the variable in Exercise 4, and state $F_X(0)$, $F_X(1)$ and $F_X(1.5)$.

6. Five components are tested independently, each failing with probability 0.3. Tabulate the pmf and CDF of the number of failures, and find $P(X \geq 2)$.

7. For the geometric variable of Example 6, find $P(N = 4)$, $P(N \leq 4)$ and $P(N > 10)$.

8. Show that for any discrete X and any $a < b$, $P(a < X \leq b) = F_X(b) - F_X(a)$, directly from the axioms.

9. A machine produces items of which 5% are defective. Let X be the number of defectives in a box of 20. Write down $p_X(x)$, and compute $P(X = 0)$ and $P(X \geq 1)$.

10. Explain why a function F that decreases anywhere cannot be a CDF, quoting the property it violates.

11. Let A and B be events and let $X = \mathbf{1}_A + \mathbf{1}_B$. Describe in words what X counts, and give $P(X = 2)$ in terms of A and B.

12. A variable takes values $1, 2, 3, \ldots$ with $p(k) = c\,2^{-k}$. Find c, then $P(X$ is even).

Answers to the odd-numbered exercises

1. On $\Omega = \{HH, HT, TH, TT\}$: $X(HH) = 2$, $X(HT) = X(TH) = 1$, $X(TT) = 0$, giving $p_X = (1/4, 1/2, 1/4)$ at $0, 1, 2$. Y counts changes: $Y(HH) = Y(TT) = 0$ and $Y(HT) = Y(TH) = 1$, so $p_Y(0) = p_Y(1) = 1/2$.

3. $c(1 + 1/2 + 1/3 + 1/4) = c(25/12) = 1$, so $c = 12/25 = 0.48$. Then $P(X \leq 2) = 0.48(1 + 0.5) = 0.72$.

5. The staircase rises at $-2, -1, 1, 2$ by $0.2, 0.1, 0.3, 0.4$. So $F_X(0) = 0.3$, $F_X(1) = 0.6$ and $F_X(1.5) = 0.6$, since nothing lies between 1 and 1.5.

7. $P(N = 4) = (0.8)^3(0.2) = 0.1024$; $P(N \leq 4) = 1 - (0.8)^4 = 0.5904$; $P(N > 10) = (0.8)^{10} = 0.1074$.

9. $p_X(x) = \binom{20}{x}(0.05)^x(0.95)^{20-x}$. Then $P(X = 0) = (0.95)^{20} = 0.3585$ and $P(X \geq 1) = 0.6415$.

11. X counts how many of the two events occur, so it takes the values $0, 1, 2$. $P(X = 2) = P(A \cap B)$, which is $P(A)P(B)$ only when A and B are independent.

Before next week

Attempt exercises 1 to 7. Week 6 takes the distributions that keep appearing, Bernoulli, binomial, geometric, negative binomial and Poisson, and asks for each one what mechanism produces it, so that you choose a distribution by argument rather than by habit.